A Constructible Tower Approaching π

Explicit Compass-and-Straightedge Approximants under Transparent Algebraic Acceleration

Wayne Baker

May 24, 2026

Abstract

We exhibit an explicit sequence of compass-and-straightedge *constructible* real numbers that approaches the geometric constant $\pi_g = \pi$, a provably non-constructible limit — under a transparent rational acceleration law. The base sequence is the Archimedean polygon $P_N = N \sin(\pi/N)$ restricted to the doubling ladder $N = 4, 8, 16, \ldots$, every term of which is a constructible length. These constructible terms are combined by the iterated N-Series recurrence with denominators $4^j - 1$, producing a triangular tableau whose diagonal entries are again constructible and which approach π_g with order increasing without bound. The point of interest is not the speed in isolation — the rate is classical Romberg behavior — but the *combination*: an explicit, field-theoretically constructible tower, accelerated by an elementary transparent recurrence, pressed against the constructibility boundary at superconvergent rate. We are careful to state what this does and does not mean for the constructible/analytic divide: the boundary is honored, not crossed.

1 The Two Facts in Tension

Two facts sit on either side of this note.

Fact 1 (geometry). For N a Gauss–Wantzel constructible order — in particular every power of two — the regular N-gon is constructible by compass and straightedge, and so is its associated polygon quantity

$$P_N = N \sin\left(\frac{\pi}{N}\right),$$

realized as a finite sequence of constructible chord operations. On the doubling ladder $N = 4, 8, 16, 32, \ldots$ every P_N is therefore a constructible real number. We write the geometric limit of this constructible construction as

$$\pi_g := \lim_{N \to \infty} P_N,$$

the N-Series *constructible circle constant*: a quantity defined purely by polygonal, scale-cancellation geometry rather than by any analytic series.

Fact 2 (analysis). The geometric limit π_g is *not* a constructible number: it coincides numerically with π, which by Lindemann's theorem is transcendental and hence lies in no finite tower of quadratic extensions of $\mathbb{Q}$. Thus π_g is *approached* by constructible quantities but is not itself constructible.

That constructible approximants approach a non-constructible limit is, by itself, ancient and trivial: Archimedes' polygons already do it. The content of this note is what happens when those constructible approximants are combined by an explicit algebraic acceleration that keeps every intermediate quantity constructible while driving the approach to superconvergent order.

2 The Acceleration Keeps the Tower Constructible

Remark 1 (No circular use of π)**.** *Although the analytic expression $P_N = N\sin(\pi/N)$ contains π, the construction of P_N does not require π as data. Geometrically, P_N is one half of the perimeter of the regular N-gon inscribed in the unit circle. For $N = 4\cdot 2^i$, this polygon is constructible by repeated bisection, and its side length is obtained by ordinary chord construction. The occurrence of π in the displayed formula is therefore an analytic* description *of the constructed length, not an ingredient used to construct it.*

Fix $k = 2$ and the doubling ladder $T_{i,0} = P(4\cdot 2^{\,i})$. The iterated N-Series operator is

$$\boxed{T_{i,j} = T_{i,j-1} + \frac{T_{i,j-1} - T_{i-1,j-1}}{4^j - 1}}$$

with denominators $4^j - 1 = 3, 15, 63, 255, \ldots$ The diagonal $T_{i,i}$ is the working estimate.

Proposition 1 (Constructibility is preserved)**.** *Every tableau entry $T_{i,j}$, and in particular every diagonal entry $T_{i,i}$, is a constructible real number.*

Proof. The base entries $T_{i,0} = P(4\cdot 2^i)$ are constructible by Fact 1, since $4\cdot 2^i$ is a power of two. The recurrence forms $T_{i,j}$ from $T_{i,j-1}$ and $T_{i-1,j-1}$ using only addition, subtraction, and division by the rational integer $4^j - 1$. The constructible reals form a field; they are closed under these operations. By induction on j, every $T_{i,j}$ is constructible. □

Thus the diagonal is an *explicit tower of constructible numbers* — each one writable, in principle, as a finite compass-and-straightedge construction followed by rational combination.

Definition 1 (The geometric circle constant)**.** *The diagonal of the constructible accelerated tower converges to the same limit as the underlying Archimedean sequence. This follows directly from the diagonal error estimate established in the next section, $|T_{i,i} - \pi| = O\big(N_i^{-2(i+1)}\big)$ with $N_i = 4\cdot 2^i$: since $N_i \to \infty$, this bound tends to zero, so the diagonal converges to π by its own error estimate (not merely as an inheritance from fixed columns, whose index does not track the diagonal). Writing this limit as π_g,*

$$\pi_g := \lim_{i\to\infty} T_{i,i} \;=\; \lim_{N\to\infty} P_N \;=\; \pi.$$

Thus π_g is not a new numerical constant; it is a constructible geometric route to the classical circle constant, with the constructible tower approaching the non-constructible value $\pi_g = \pi$.

3 The Rate (Stated Honestly)

The base sequence has the even-power expansion $P_N = \pi + C_2N^{-2} + C_4N^{-4} + \cdots$. Column j of the tableau annihilates the $O(N^{-2j})$ term, so the diagonal satisfies

$$T_{i,i} = \pi + O\big(N^{-2(i+1)}\big), \qquad N = 4\cdot 2^i,$$

order increasing by two at each level. Writing $D(i) = -\log_{10}|T_{i,i} - \pi|$ for the correct-digit count, the leading behavior is, with an honest error term,

$$D(i) = 2\log_{10}(2)\, i^2 + O(i), \qquad \text{equivalently} \qquad \frac{D(i)}{i^2} \longrightarrow 2\log_{10}(2) \approx 0.602.$$

Remark 2 (Finite-level behavior)**.** *The $O(i)$ correction is negative and non-negligible at accessible levels: the measured ratio $D(i)/i^2$ in Table 1 descends through the data and has not reached the* 0.602 *leading coefficient by level 30 (it reads ≈ 0.415 there and is still falling slowly). The quadratic-in-level growth is the honest description of the leading term; we do* not *claim the finite-level coefficient has stabilized.*

Level i	$N = 4 \cdot 2^i$	Correct digits $D(i)$	$D(i)/i^2$
5	128	19.59	0.78355
10	4096	57.34	0.57337
15	131072	112.05	0.49798
20	4194304	183.12	0.45781
25	134217728	270.26	0.43241
30	4294967296	373.26	0.41474

Table 1: Diagonal of the constructible tower (mpmath, 600 digits). Correct digits grow quadratically in level; the ratio column is still descending toward the asymptotic $2\log_{10} 2$ and is reported as-measured, not extrapolated.

4 What This Means — and What It Does Not

The Claim

There exists an explicit sequence of compass-and-straightedge constructible real numbers, generated from the Archimedean doubling-ladder polygon by the transparent rational recurrence $T_{i,j} = T_{i,j-1} + (T_{i,j-1} - T_{i-1,j-1})/(4^j - 1)$, whose accelerated diagonal defines the geometric circle constant

$$\pi_g := \lim_{i\to\infty} T_{i,i} = \pi,$$

where every finite approximant $T_{i,i}$ remains constructible although the limit $\pi_g = \pi$ is transcendental. The diagonal converges with order of accuracy increasing without bound (quadratic-in-level digit growth, $D(i)/i^2 \to 2\log_{10} 2$). The novelty is the combination of full explicit constructibility with transparent algebraic superconvergence toward the constructibility boundary, with π_g named as the geometric-limit route to the classical constant rather than as a new number.

What Is Not Claimed

This note does *not* claim that π is constructible (it is transcendental, by Lindemann), that the constructible/analytic boundary is crossed or dissolved (it is honored: every approximant stays strictly inside the constructible field, the limit stays strictly outside it), that the method competes with modular-form series in arithmetic cost, that the acceleration is novel as a convergence technique (it is classical Richardson/Romberg applied to an even-power sequence), or any result outside the tested doubling-ladder construction. The acceleration moves the approximants toward the boundary faster; it does not move the boundary.

5 Why It May Earn a Second Look

Convergence-rate notes on π are plentiful and this one would vanish among them if pitched as a speed result. The distinguishing feature is provenance and explicitness: the entire tower lives inside the constructible field, is built from a literal compass-and-straightedge sequence, and is accelerated by a recurrence whose only constants are the plainly visible $4^j - 1$. It is, in effect, a precise *survey of the edge of constructibility* — a marked, explicit approach to a boundary that Wantzel's and Lindemann's theorems guarantee cannot be reached. The interest is geometric and structural, not computational.

Closing Identity

The construction is geometric throughout: the approximants are constructible, the recurrence is rational and transparent, and the limit is the geometric constant π_g. Numerically, to all tested precision (here 600 digits, 373 correct at level 30),

$$\pi_g \;=\; \pi.$$

The note thus describes a constructible geometric tower whose limiting value is $\pi_g = \pi$, numerically confirmed here to the tested precision, while the limit itself remains non-constructible.